Nordwerk Schaltanlagen GmbH

Technical and Commercial Offer

Product: NWS-36 GIS

Submitted in response to Material Requisition MR-33kV-GIS-Rev4

1. COVER LETTER

This offer remains valid for the period stated in the commercial terms.

Our offer is in full compliance with MR-33kV-GIS-Rev4 except minor commercial notes.

Yours faithfully,

Dear Sirs,

Nordwerk Schaltanlagen GmbH

2. COMPLIANCE STATEMENT

Rated short-time withstand current - Complied, see datasheet

Busbar material and rating - Complied, see datasheet

Enclosure and internal arc classification - Complied, see datasheet

Protection and SCADA communication - Complied, see datasheet

Factory acceptance testing - Complied, see datasheet

Delivery terms and warranty - Complied, see datasheet

Rated voltage - Complied, see datasheet

Ambient temperature rating - Complied, see datasheet

Information provided by the purchaser is for tender purposes and remains confidential.

The supplier shall comply with the purchaser's site rules and permit-to-work procedures.

This section describes the general obligations of the supplier for the scope defined in this document.

3. TECHNICAL DATASHEET

The datasheet below is completed against the purchaser's datasheet template.

Installation: indoor, in a prefabricated substation building.

Insulating medium: fluorine-free gas mixture.

Operating mechanism: motor-charged spring mechanism.

Installation: indoor, in a prefabricated substation building.

Rated short-time withstand current (main busbar and breaker): 50 kA / 3 s.

Main busbars: silver-plated high-conductivity electrolytic copper Cu-ETP, 2.5 kA continuous.

Rated voltage Ur: 36 kV.

Insulating medium: fluorine-free gas mixture.

Operating mechanism: motor-charged spring mechanism.

Insulating medium: fluorine-free gas mixture.

Enclosure protection IP54 with rear gas exhaust ducting, internal arc classification IAC AFLR 50 kA 1 s.

Rated for ambient temperatures from -25 °C to +40 °C without derating.

Operating mechanism: motor-charged spring mechanism.

Mechanical endurance class according to the applicable standard.

Operating mechanism: motor-charged spring mechanism.

Protection IEDs NW-Prot 5 with IEC 61850 Ed.2 PRP dual-port fibre, pre-wired to the central RTU marshalling cabinet.

Mechanical endurance class according to the applicable standard.

The datasheet below is completed against the purchaser's datasheet template.

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Installation: indoor, in a prefabricated substation building.

Insulating medium: fluorine-free gas mixture.

4. SINGLE LINE DIAGRAM NOTES

NOTE 4: Cable terminations are of the plug-in type.

NOTE 5: The bus coupler connects both switchgear sections.

NOTE 6: Auxiliary supplies are distributed from the station board.

NOTE 8: Refer to the general arrangement drawing for panel positions.

Drawing NWS-36-SLD - sheet 1

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NOTE 6: Auxiliary supplies are distributed from the station board.

NOTE 8: Refer to the general arrangement drawing for panel positions.

NOTE 1: All dimensions are in millimetres unless stated otherwise.

Drawing NWS-36-SLD - sheet 2

NOTE 6: Auxiliary supplies are distributed from the station board.

NOTE 8: Refer to the general arrangement drawing for panel positions.

NOTE 1: All dimensions are in millimetres unless stated otherwise.

NOTE 2: Earthing switches are interlocked with the associated circuit breakers.

Drawing NWS-36-SLD - sheet 3

NOTE 8: Refer to the general arrangement drawing for panel positions.

NOTE 1: All dimensions are in millimetres unless stated otherwise.

NOTE 2: Earthing switches are interlocked with the associated circuit breakers.

NOTE 3: Voltage transformers are fitted on incomer bays only.

Drawing NWS-36-SLD - sheet 4

5. BILL OF MATERIALS

Pos 10: Circuit breaker panels with vacuum interrupters.

Pos 12: Current transformers for protection and metering.

Pos 20: Cable termination compartments.

Pos 30: Low voltage compartments with terminal blocks.

Pos 12: Current transformers for protection and metering.

Pos 20: Cable termination compartments.

Pos 30: Low voltage compartments with terminal blocks.

Pos 40: Protection relay assemblies.

Pos 20: Cable termination compartments.

Pos 30: Low voltage compartments with terminal blocks.

Pos 40: Protection relay assemblies.

Pos 50: Gas monitoring devices.

6. INSPECTION AND TEST PLAN

Type test certificates from an accredited laboratory are submitted with the offer.

Test results are recorded on test sheets signed by the supplier's quality inspector.

Hold points are agreed with the purchaser during the kick-off meeting.

The inspection and test plan lists all inspections and tests during manufacturing.

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The price includes a 5-day client-witnessed I-FAT with primary injection.

Hold points are agreed with the purchaser during the kick-off meeting.

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The inspection and test plan lists all inspections and tests during manufacturing.

Type test certificates from an accredited laboratory are submitted with the offer.

Test results are recorded on test sheets signed by the supplier's quality inspector.

7. COMMERCIAL TERMS AND PRICE SCHEDULE

Prices are fixed and firm for the scope described in this offer.

Payment terms: milestone payments as defined in the purchase order.

The offer is subject to the purchaser's general purchasing conditions.

Taxes are shown separately where applicable.

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Delivery DDP project site Brandenburg (Incoterms 2020) with a 36-month defect liability period.

The offer is subject to the purchaser's general purchasing conditions.

Taxes are shown separately where applicable.

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Total lump-sum price for the complete scope: €452,000

Taxes are shown separately where applicable.

Unit prices for additional bays are available on request.

8. QUALITY ASSURANCE

The purchaser reserves the right to audit the supplier and its sub-suppliers.

Material certificates shall be traceable to the delivered equipment.

The supplier shall operate a quality management system certified to ISO 9001.

A quality plan shall be issued for review before manufacturing starts.

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Non-conformances shall be reported to the purchaser together with the proposed corrective action.

9. SPARE PARTS

Spare parts shall be interchangeable with the corresponding original parts.

Special tools required for maintenance shall be supplied with the equipment.

Spare part prices shall remain valid as stated in the commercial terms.

A list of recommended spare parts shall be included with the offer.

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Spare parts shall be interchangeable with the corresponding original parts.

10. TRAINING

Training shall cover operation, maintenance and safety of the switchgear.

Training material shall be provided to each participant.

The training programme shall be agreed with the purchaser before delivery.

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11. DOCUMENTATION

Document reviews by the purchaser do not relieve the supplier of responsibility for correctness.

Drawings and documents shall be submitted in electronic format for review and approval.

Each document shall carry a title block with document number, revision and approval status.

As-built documentation shall be delivered after commissioning is complete.

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Operation and maintenance manuals shall be provided in English and German.